

ElderGuard AI — Technical Progress Sheet

Project: ElderGuard AI | **Track:** Track A (IoT Product & System Architecture)
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1. Problem Statement

Independent seniors face high risks from unnoticed falls, abnormal vitals, and missed medications. When admitted to clinics, doctors lack immediate records of recent vital trends or medication history, delaying urgent care.

2. Proposed Solution

An IoT healthcare ecosystem combining a dual-mode Flutter app (Elderly/Caregiver) and a zero-login clinical QR web dashboard. An ESP32 wearable streams vitals and motion via BLE to the elderly phone for local reminders and risk checks. Critical emergencies trigger instant caregiver notifications. Clinic staff scan a unique QR code on the patient's phone to instantly view and print a 1-page summary report (Ctrl + P) for the consulting physician.

3. System Architecture & Data Flow

ESP32 Wearable (Vitals/IMU) → [BLE] → Flutter App → [HTTPS] → FastAPI + MySQL → Caregiver Push Alert / Hospital Web (Ctrl+P)

4. Current Implementation Progress

- **Figma UI/UX:** Complete interactive flows for Elderly Mode (high-contrast vitals/QR) and Caregiver Mode (remote trends/alerts).
- **FastAPI Backend:** Functional REST endpoints for telemetry ingestion, tokenized patient IDs, and multi-tier risk evaluation.
- **Clinical Web Dashboard:** Web viewer built with a dedicated @media print CSS stylesheet for 1-click clean physical printing.
- **Firmware Logic:** Sensor thresholds, data formatting, and fall-detection algorithms validated via Wokwi ESP32 simulation.

5. Technology Stack

Layer	Selected Technologies
IoT / Hardware	ESP32, MAX30102 (SpO ₂ / HR), DS18B20 (Temp), MPU6050 (Motion / Fall)
Protocols	Bluetooth Low Energy (BLE) & HTTPS / REST
Software & Web	Flutter / Dart (Mobile), FastAPI / Python (Backend), MySQL (Database), Tailwind CSS (Web)

6. Testing & Validation

- **Risk Engine:** Simulated telemetry packets passed to FastAPI; critical thresholds (e.g., HR > 120 bpm + fall event) trigger alert statuses in under 120 ms.
- **Print Validation:** Tested tokenized QR URLs on major browsers; Ctrl + P generates a properly formatted 1-page physical report with zero margin clipping.

7. Limitations & Technical Risks

- **Hardware Setup:** Currently tested via simulated ESP32 streams; physical circuit assembly transitions during the Grand Finale phase.
- **False Triggers:** Motion spikes can trigger false falls; mitigated by a 30-second mobile cancellation countdown before alerting caregivers.

8. Next Steps (Roadmap for Grand Finale – Oct 17)

- Flash physical ESP32 and pair hardware sensors via native BLE.
- Deploy backend and web dashboard to cloud infrastructure (AWS / Render).
- Perform end-to-end physical drop tests and live vital sign demonstration.